INSTITUTO TECNOLÓGICO Y DE ESTUDIOS

SUPERIORES DE MONTERREY

Actividad 4.2. Ejercicio de programación 1

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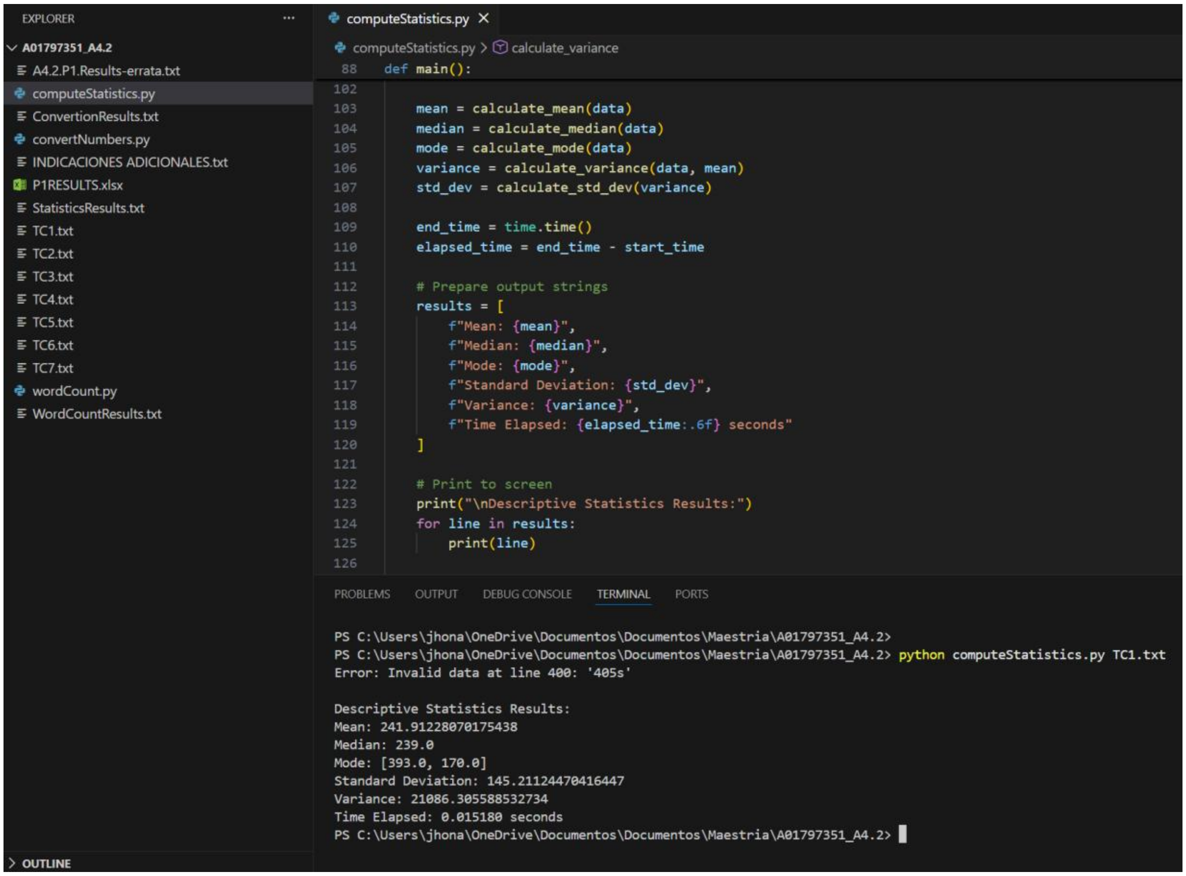
Pruebas de software y aseguramiento de la calidad (Gpo 10)

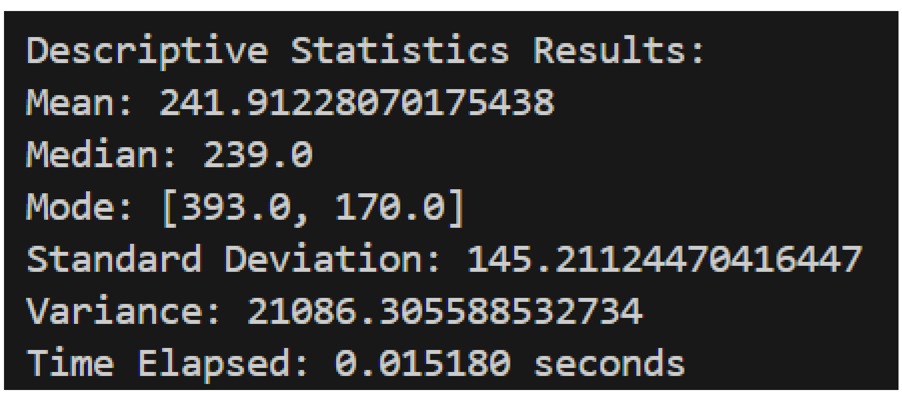
08 de febrero de 2026

1. Ejercicio Compute statistics, resultados documentados
   Evidencia de ejecución exitosa con el archivo de prueba TC1.txt. Se muestran los cálculos de media, mediana, moda, desviación estándar y varianza poblacional, así como el tiempo de ejecución.

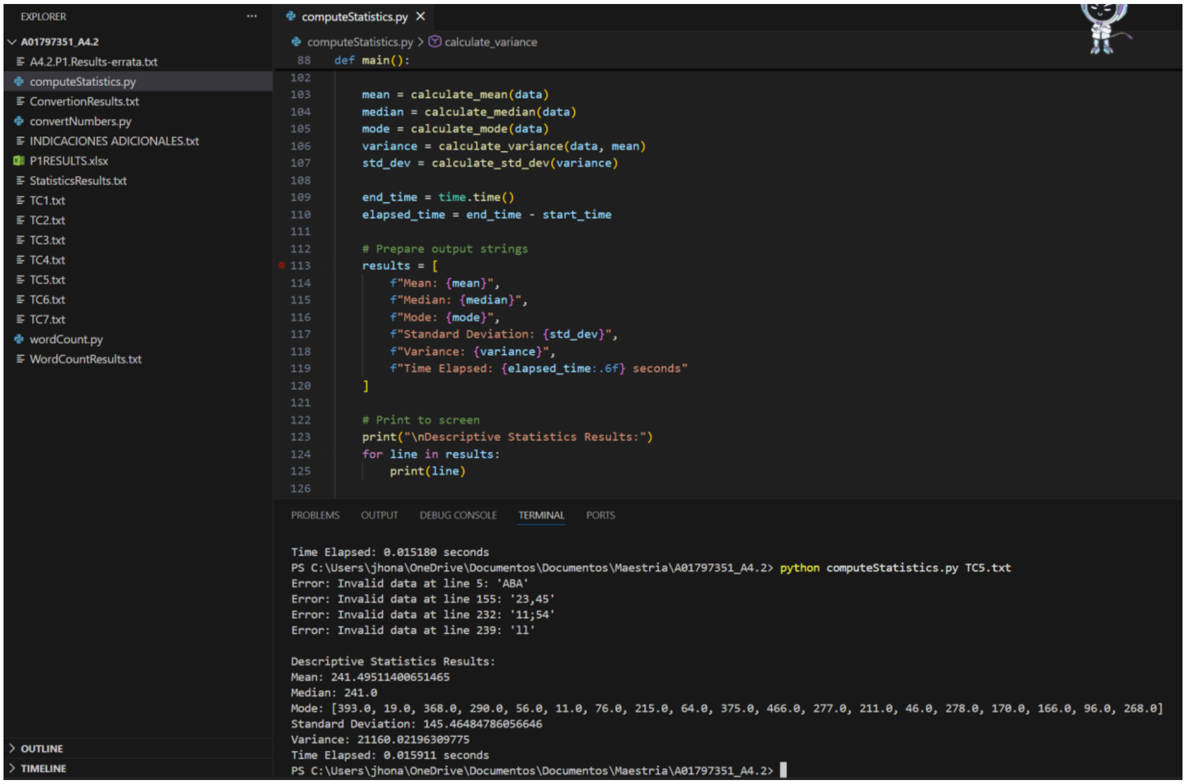



```
def main():

    mean = calculate_mean(data)
    median = calculate_median(data)
    mode = calculate_mode(data)
    variance = calculate_variance(data, mean)
    std_dev = calculate_std_dev(variance)

    end_time = time.time()
    elapsed_time = end_time - start_time

    # Prepare output strings
    results = [
        f"Mean: {mean}",
        f"Median: {median}",
        f"Mode: {mode}",
        f"Standard Deviation: {std_dev}",
        f"Variance: {variance}",
        f"Time Elapsed: {elapsed_time:.6f} seconds"
    ]

    # Print to screen
    print("\nDescriptive Statistics Results:")
    for line in results:
        print(line)
```

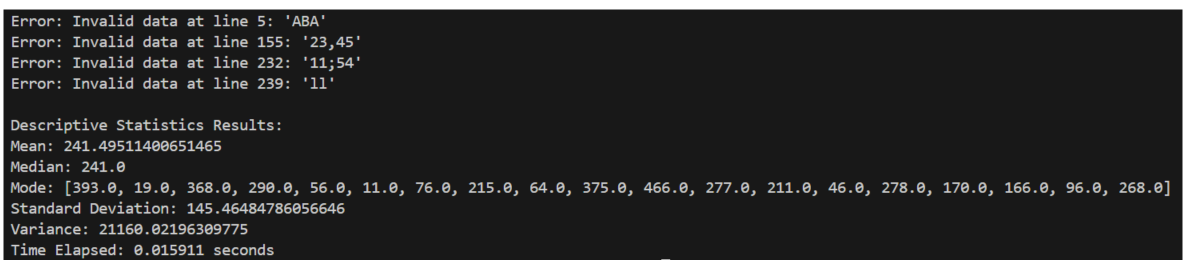
```
PS C:\Users\jhona\OneDrive\Documentos\Documentos\Maestria\A01797351_A4.2>
PS C:\Users\jhona\OneDrive\Documentos\Documentos\Maestria\A01797351_A4.2> python computeStatistics.py TC1.txt
Error: Invalid data at line 400: '405s'

Descriptive Statistics Results:
Mean: 241.91228070175438
Median: 239.0
Mode: [393.0, 170.0]
Standard Deviation: 145.21124470416447
Variance: 21086.305588532734
Time Elapsed: 0.015180 seconds
PS C:\Users\jhona\OneDrive\Documentos\Documentos\Maestria\A01797351_A4.2>
```



```
Descriptive Statistics Results:
Mean: 241.91228070175438
Median: 239.0
Mode: [393.0, 170.0]
Standard Deviation: 145.21124470416447
Variance: 21086.305588532734
Time Elapsed: 0.015180 seconds
```

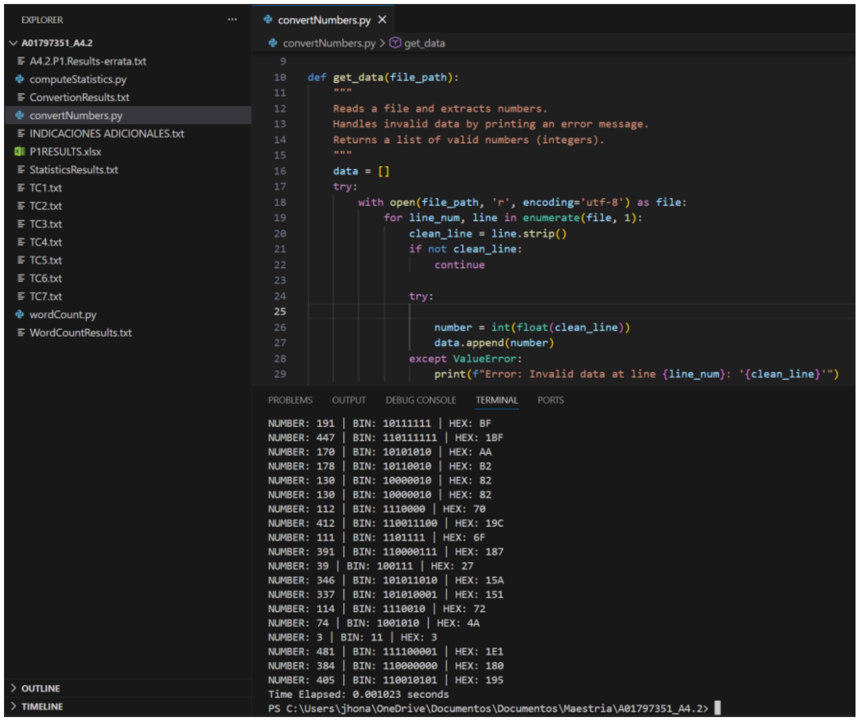


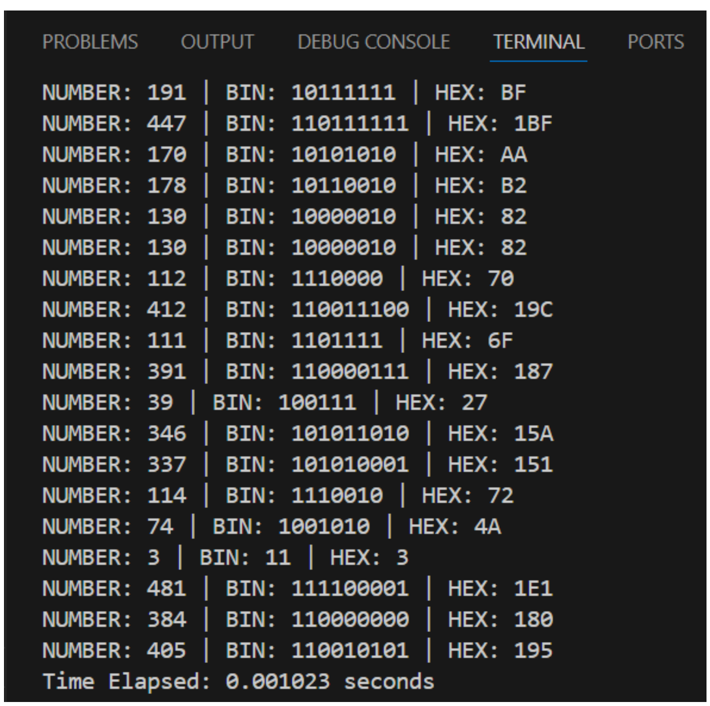
Prueba de manejo de errores con el archivo TC5.txt. El programa identifica el dato inválido ABA, muestra el error en consola y continúa la ejecución correctamente.

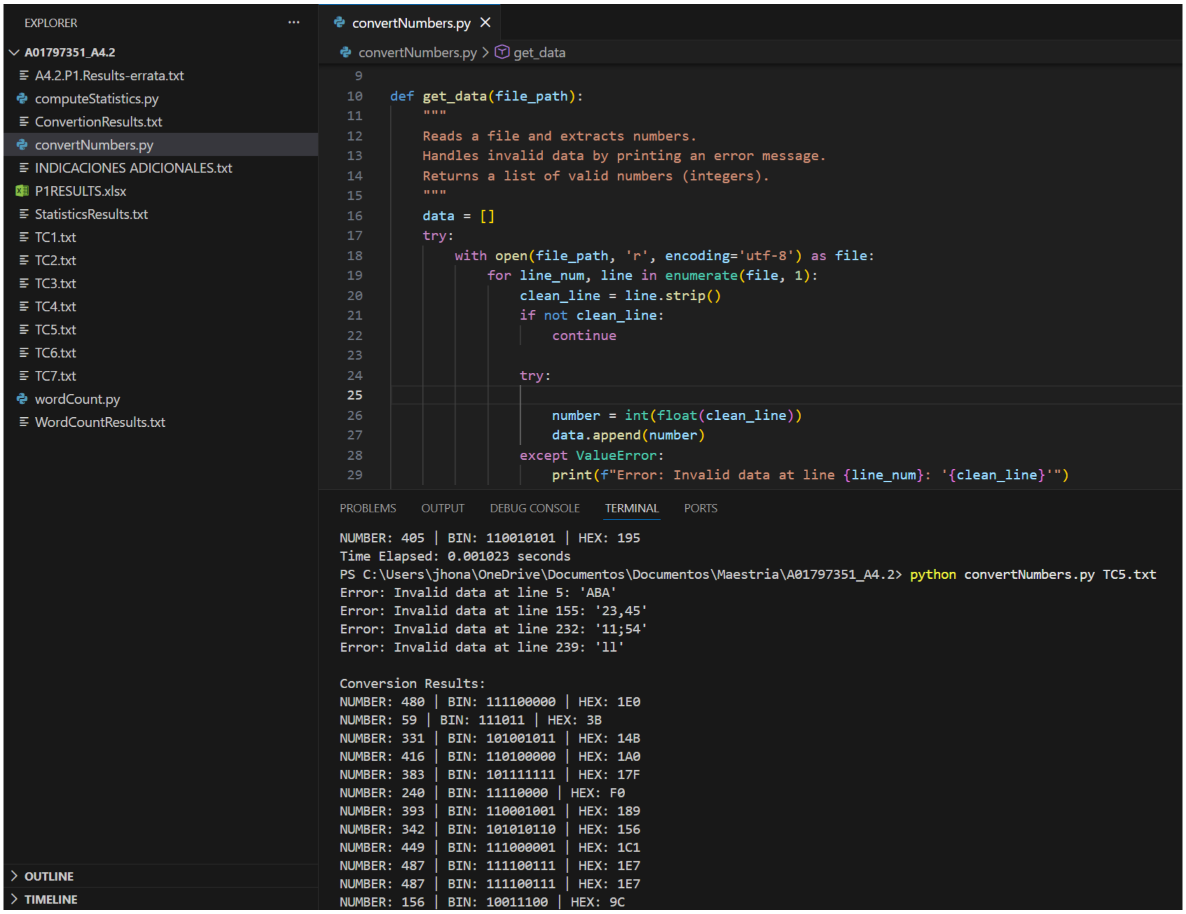


```
Error: Invalid data at line 5: 'ABA'
Error: Invalid data at line 155: '23,45'
Error: Invalid data at line 232: '11;54'
Error: Invalid data at line 239: '1l'

Descriptive Statistics Results:
Mean: 241.49511400651465
Median: 241.0
Mode: [393.0, 19.0, 368.0, 290.0, 56.0, 11.0, 76.0, 215.0, 64.0, 375.0, 466.0, 277.0, 211.0, 46.0, 278.0, 170.0, 166.0, 96.0, 268.0]
Standard Deviation: 145.46484786056646
Variance: 21160.02196309775
Time Elapsed: 0.015911 seconds
```

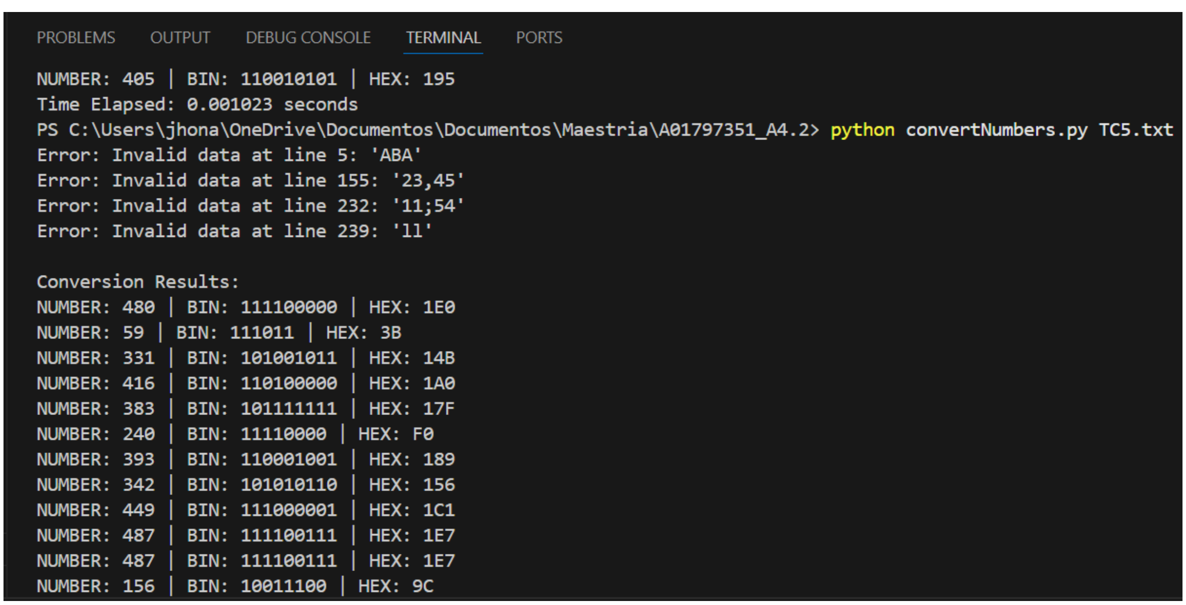
2. Ejercicio Converter, resultados documentados.
   Conversión correcta de números a bases binaria y hexadecimal utilizando el archivo de prueba TC1.txt.







El sistema maneja correctamente datos no numéricos imprimiendo la advertencia correspondiente sin detener el programa.



```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

NUMBER: 3366082426106749911O4 | BIN: 100100011111101100000101101010011101000111101110010100000000000000000 | HEX: 123F60B53A3DCA0000
NUMBER: 3575605394566350110 72 | BIN: 100110110001000100110001110000011001011101001110001100000000000000000 | HEX: 1362263832E9C60000
NUMBER: 2519532937874709872 64 | BIN: 110110101000100011011110100111001011001111111101011110000000000000000 | HEX: DA88DE9CB3FD78000
NUMBER: 6049789800016349 5936 | BIN: 1101000111100100111011010110110011000101011000111101100000000000000 | HEX: 34793B5B3158F6000
NUMBER: 10159168112590900 4288 | BIN: 101100000001110111100010011110111010011011001001000000000000000000000 | HEX: 581DE27BA6C900000
NUMBER: 30482969917149400 2688 | BIN: 10000100001100101110010100100010101101110101000010110000000000000000 | HEX: 10865CA456EA160000
NUMBER: 24212457494373698 7648 | BIN: 1101001000000010011101001010000101001001000111100011100000000000000000 | HEX: D20274A1491E38000
NUMBER: 24106536737521100 3904 | BIN: 110100010001011101000011101001101000101000000111111000000000000000000 | HEX: D11743A68A07E0000
NUMBER: 17248040195625500 6720 | BIN: 100101011001101001010111101110111101001001110010101110000000000000000 | HEX: 959A57BBD272B8000
NUMBER: 31818604709094300 0576 | BIN: 10001001111111011011111101100000000111011101010111110000000000000000000 | HEX: 113FB7EC01DD5F0000
NUMBER: 9513367755441849 9584 | BIN: 1010010100000111110101101111011001000100111101010101100000000000000000 | HEX: 5283EB7B227AAC000
NUMBER: 15505179641585300 2752 | BIN: 100001100111110001101010001101010111001100010001101010000000000000000 | HEX: 867C6A357311A8000
NUMBER: 9784627237677100 2368 | BIN: 101010011011111000111100101001010000100111011000110101000000000000000 | HEX: 54DE3CA509D8D4000
NUMBER: 12669625625850300 0064 | BIN: 110110111100100001110000101110011010011001111101110110000000000000000 | HEX: 6DE4385CD33EEC000
NUMBER: 1015896100216009 9328 | BIN: 1000110011111011111000010011010011111110011001100100100000000000000 | HEX: 8CFBE134FCCC9000
NUMBER: 7068011770755600 3840 | BIN: 1111010100111000100011100010010011011000011010110000100000000000000 | HEX: 3D4E2389361AC2000
NUMBER: 9403939149936980 7872 | BIN: 1010001100100001111000010000101001110011101110100111000000000000000000 | HEX: 5190F08539DD38000
NUMBER: 23328458369904200 9088 | BIN: 110010100101011110010101010011100001001011000100100000000000000000000 | HEX: CA57954E12C480000
NUMBER: 6036487663253169 7664 | BIN: 1101000101101110110001111101111000011100000101010010000000000000000000 | HEX: 345BB1F7870548000
Time Elapsed: 0.072619 seconds
PS C:\Users\jhona\OneDrive\Documentos\Documentos\Maestria\A01797351_A4.2>
```

```
EXPLORER
A01797351_A4.2
  A4.2.P1.Results-errata.txt
  computeStatistics.py
  ConvertionResults.txt
  convertNumbers.py
  INDICACIONES ADICIONALES.txt
  P1RESULTS.xlsx
  StatisticsResults.txt
  TC1.txt
  TC2.txt
  TC3.txt
  TC4.txt
  TC5.txt
  TC6.txt
  TC7.txt
  wordCount.py
  WordCountResults.txt
OUTLINE
TIMELINE
```

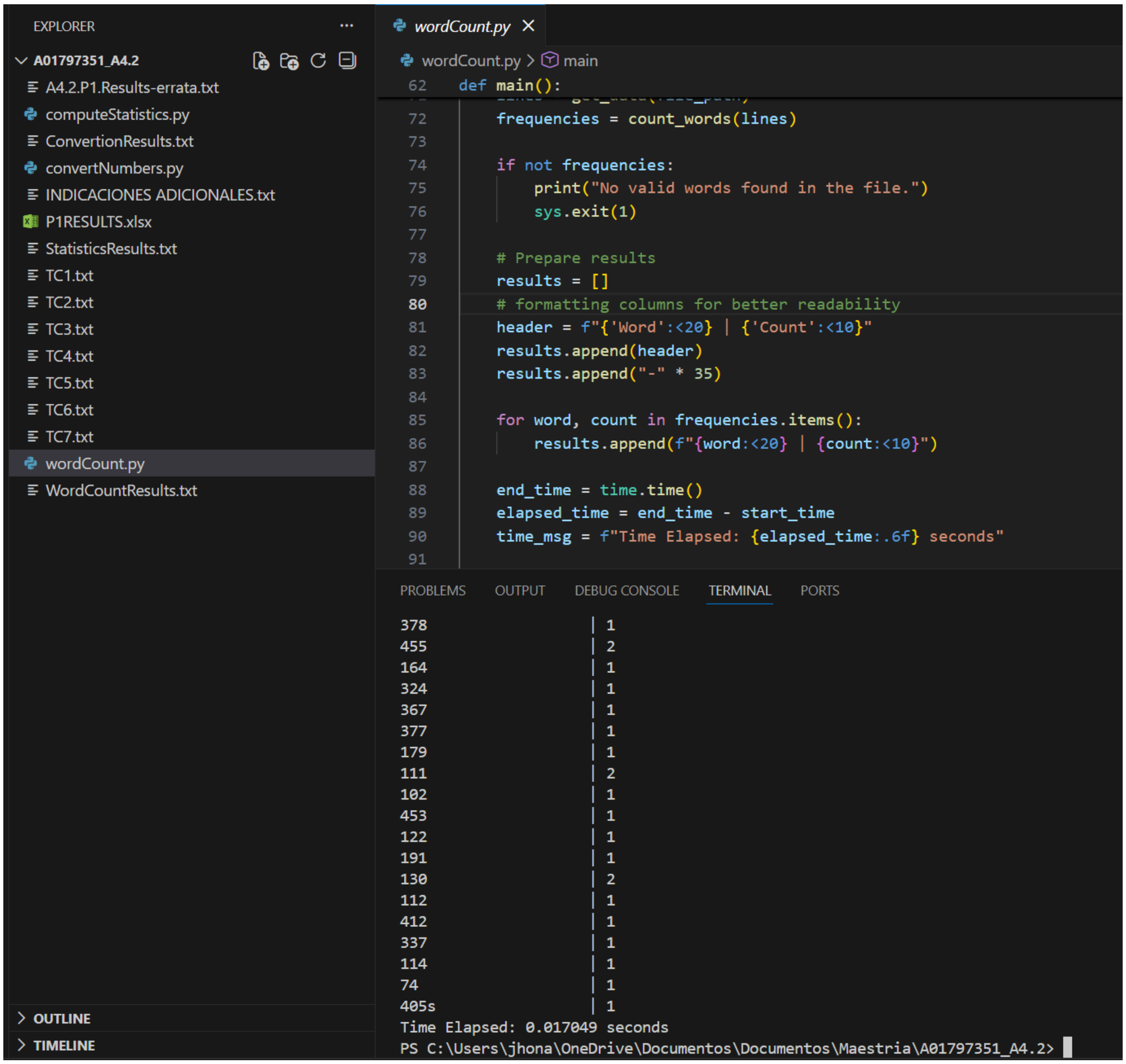
```python
# convertNumbers.py > get_data

def get_data(file_path):
    """
    Reads a file and extracts numbers.
    Handles invalid data by printing an error message.
    Returns a list of valid numbers (integers).
    """
    data = []
    try:
        with open(file_path, 'r', encoding='utf-8') as file:
            for line_num, line in enumerate(file, 1):
                clean_line = line.strip()
                if not clean_line:
                    continue

                try:

                    number = int(float(clean_line))
                    data.append(number)
                except ValueError:
                    print(f"Error: Invalid data at line {line_num}: '{clean_line}'")
```

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

NUMBER: 331 | BIN: 101001011 | HEX: 14B
NUMBER: 268 | BIN: 100001100 | HEX: 10C
NUMBER: 390 | BIN: 110000110 | HEX: 186
NUMBER: 282 | BIN: 100011010 | HEX: 11A
NUMBER: 63 | BIN: 111111 | HEX: 3F
NUMBER: 332 | BIN: 101001100 | HEX: 14C
NUMBER: 211 | BIN: 11010011 | HEX: D3
NUMBER: 109 | BIN: 1101101 | HEX: 6D
NUMBER: 385 | BIN: 110000001 | HEX: 181
NUMBER: 27 | BIN: 11011 | HEX: 1B
Time Elapsed: 0.001205 seconds
PS C:\Users\jhona\OneDrive\Documentos\Documentos\Maestria\A01797351_A4.2> pylint convertNumbers.py
************* Module convertNumbers
convertNumbers.py:25:0: C0303: Trailing whitespace (trailing-whitespace)
convertNumbers.py:126:0: C0304: Final newline missing (missing-final-newline)
convertNumbers.py:1:0: C0103: Module name "convertNumbers" doesn't conform to snake_case naming style (invalid-name)

------------------------------------------------------------------
Your code has been rated at 9.61/10 (previous run: 9.74/10, -0.13)

PS C:\Users\jhona\OneDrive\Documentos\Documentos\Maestria\A01797351_A4.2>
```

Validación de estilo PEP-8 con Pylint: 10.00/10.


```
************* Module convertNumbers
convertNumbers.py:25:0: C0303: Trailing whitespace (trailing-whitespace)
convertNumbers.py:126:0: C0304: Final newline missing (missing-final-newline)
convertNumbers.py:1:0: C0103: Module name "convertNumbers" doesn't conform to snake_case naming style (invalid-name)
```

3. Ejercicio Count Words resultados documentados
   Conteo de frecuencia de palabras ejecutado con el archivo de prueba. Se visualizan las palabras y su número de apariciones



Validación de estilo PEP-8 con Pylint: 10.00/10

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
102                 | 1
453                 | 1
122                 | 1
191                 | 1
130                 | 2
112                 | 1
412                 | 1
337                 | 1
114                 | 1
74                  | 1
405s                | 1
Time Elapsed: 0.017049 seconds
PS C:\Users\jhona\OneDrive\Documentos\Documentos\Maestria\A01797351_A4.2> pylint wordCount.py
************* Module wordCount
wordCount.py:107:0: C0304: Final newline missing (missing-final-newline)
wordCount.py:1:0: C0103: Module name "wordCount" doesn't conform to snake_case naming style (invalid-name)

------------------------------------------------------------------
Your code has been rated at 9.68/10 (previous run: 9.68/10, +0.00)

PS C:\Users\jhona\OneDrive\Documentos\Documentos\Maestria\A01797351_A4.2>
```